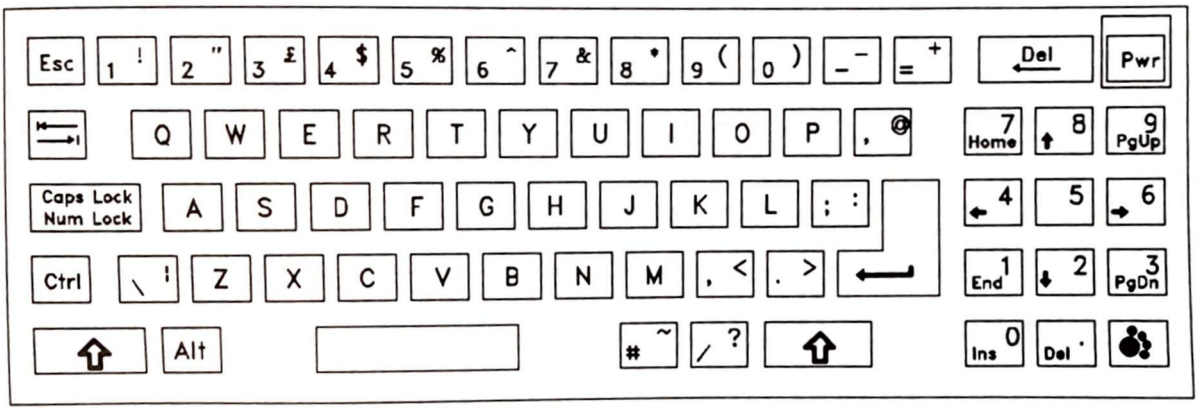
Appendix E
Keyboards

E.1 Keyboard layouts

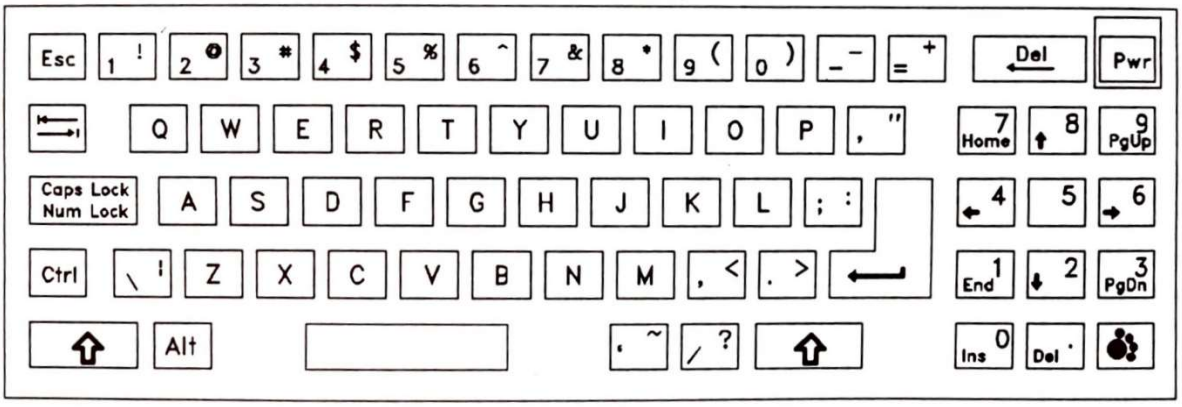
The standard UK keyboard layout is as follows:



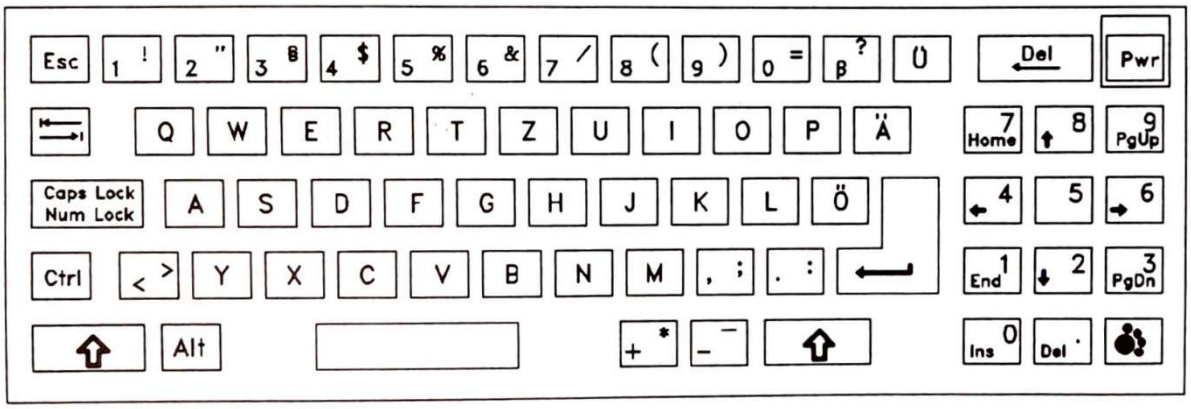
The keyboard may also be redefined by using special **Hunter 16** functions (see Part 3).

To cater for international keyboards, use the INTKEYB utility described in Part 2, not the KEYB command described in the *MS-DOS User's Reference*. A range of international keyboard layouts are available:

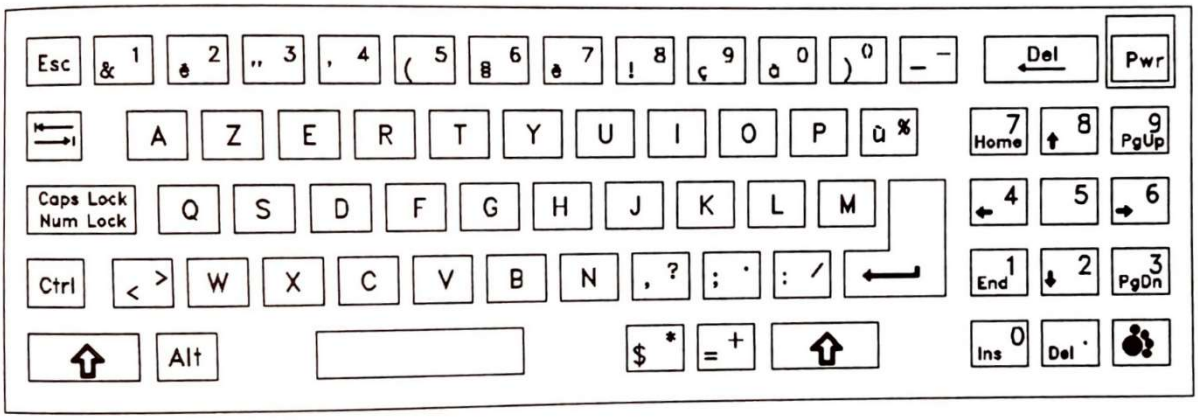
US American Keyboard



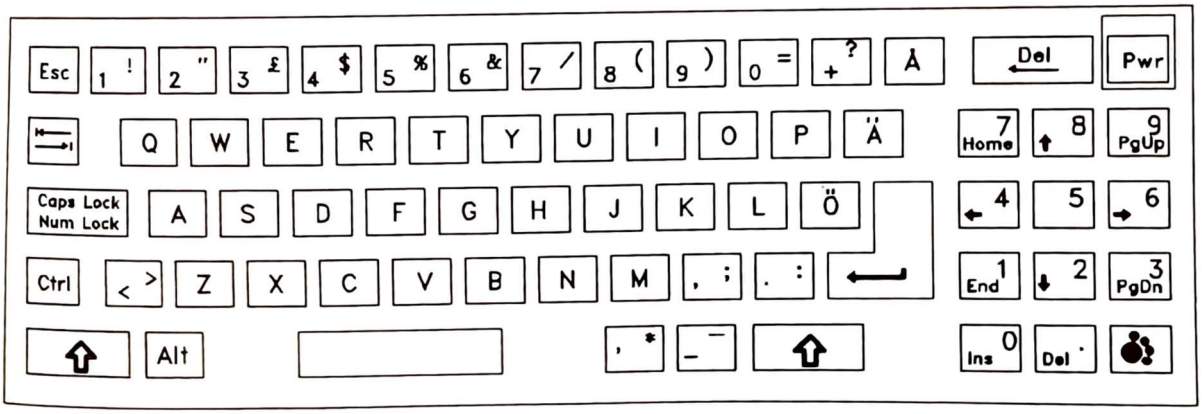
German Keyboard Layout



French Keyboard Layout



Swedish Keyboard Layout



E.2 Key matrix codes

The **Hunter 16** keyboard can be reprogrammed at a low level, by changing the key code assigned to each physical key, using INT 16H function 24H. By using this facility, you can reprogram any key on keyboard to act as any other key. The key matrix codes, which are used to identify the physical keys, are as follows:

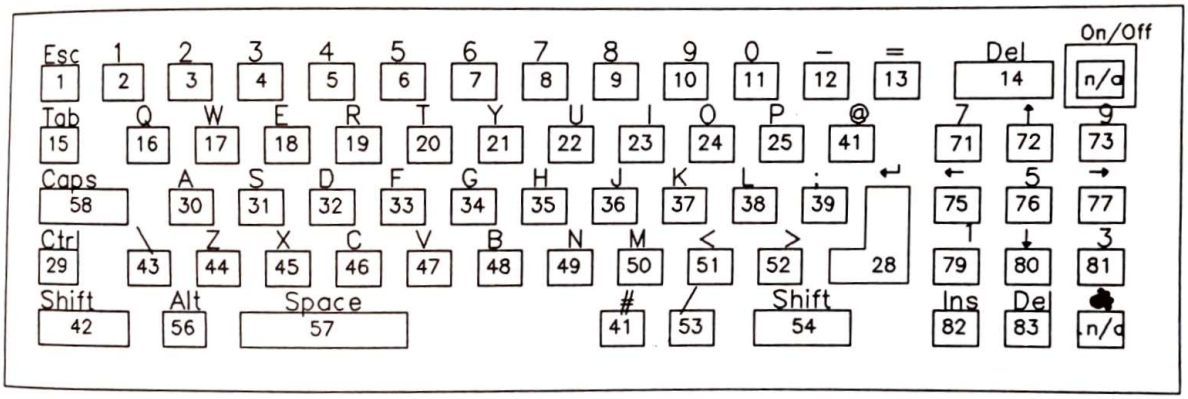
Key matrix codes

Code	Key	Code	Key	Code	Key
0	[ESC]	30	[space]	60	[.]
1	[1]	31		61	
2	[Q]	32		62	Right [Shift]
3	[TAB right]	33	[0]	63	
4	[Num][Lock]	34	[-]	64	
5	[\]	35	[‘]	65	
6		36	keypad [7]	66	[8]
7		37	[RETURN]	67	[7]
8	Left [Shift]	38	keypad [7]	68	[U]
9	[Ctrl]	39		69	[I]
10	[PAW]	40	keypad [1]	70	[J]
11	[2]	41	keypad [0]	71	[K]
12	[W]	42		72	[M]
13	[A]	43		73	[N]
14	[S]	44	[=]	74	[/]
15	[Z]	45	[TAB left]	75	
16		46	keypad [8]	76	
17	[Alt]	47	keypad [9]	77	[6]
18		48	keypad [5]	78	[5]
19		49	keypad [6]	79	[T]
20		50	keypad [2]	80	[Y]
21		51	keypad [3]	81	[G]
22	[4]	52	keypad [.]	82	[H]
23	[3]	53		83	[B]
24	[E]	54		84	[V]
25	[R]	55	[9]	85	[#]
26	[D]	56	[O]	86	
27	[F]	57	[P]	87	
28	[X]	58	[;]	88	[PWR]
29	[C]	59	[L]	89	[Shift][PWR]

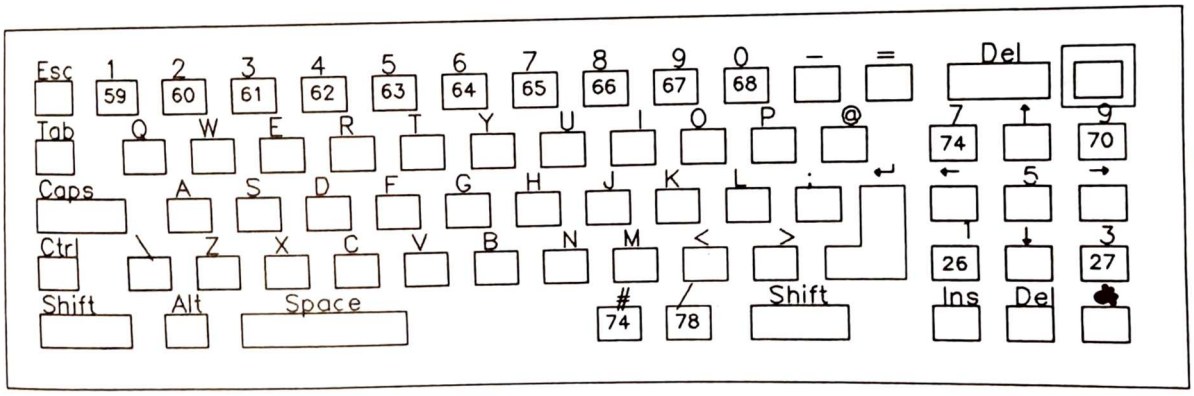
Note: there is an original text typo in code 36 or 38 – one should be a [4]

E.3 Scan codes

The **Hunter 16** keyboard provides entirely standard (IBM PC-compatible), key codes to the key scan routine (INT 9 service routine) which translates them into entirely standard scan codes. The main keys and key codes are as follows:



Certain IBM keys are typed by holding down [PAW] and pressing other keys. This generates the normal key code for that IBM key:



Note that the special Hunter 16 keys (PWR) and [PAW]) and key sequences that are used to control **Hunter 16** (such as the HOT key, LCD map and flip keys) also generate key codes, which are in addition to those generated by keys on an IBM keyboard.

Key Code	Key	Normal token	Alt token	Ctrl token	Shift token	Num lock
01	[Esc]	011B	ignore	011B	011B	-
02	[1]	0231	7800	ignore	0221	-
03	[2]	0332	7900	0300	0340	-
04	[3]	0433	7A00	ignore	0423	-
05	[4]	0534	7B00	ignore	0524	-
06	[5]	0635	7C00	ignore	0625	-
07	[6]	0736	7D00	071E	075E	-
08	[7]	0837	7E00	ignore	0826	-
09	[8]	0938	7F00	ignore	092A	-
0A	[9]	0A39	8000	ignore	0A28	-
0B	[0]	0B30	8100	ignore	0B29	-
0C	[-]	0C2D	8200	0C1F	0C5F	-
0D	[=]	0D3D	8300	ignore	0D2B	-
0E	[Tab left]	0E08	ignore	0E7F	0E08	-
0F	[Tab right]	0F09	ignore	ignore	0F00	-
10	[Q]	1071	1000	1011	1051	-
11	[W]	1177	1100	1117	1157	-
12	[E]	1265	1200	1205	1245	-
13	[R]	1372	1300	1312	1352	-
14	[T]	1474	1400	1414	1454	-
15	[Y]	1579	1500	1519	1559	-
16	[U]	1675	1600	1615	1655	-
17	[I]	1769	1700	1709	1749	-
18	[O]	186F	1800	180F	184F	-
19	[P]	1970	1900	1910	1950	-
1A	[[]	1A5B	ignore	1A1B	1A7B	-
1B	[]]	1B5D	ignore	1B1D	1B7D	-
1C	[RETURN]	1C0D	ignore	1C0A	1C0D	-
1D	[Ctrl]	ignore	ignore	-	ignore	-
1E	[A]	1E61	1E00	1E01	1E41	-
1F	[S]	1F73	1F00	1F13	1F53	-
20	[D]	2064	2000	2004	2044	-

Key Code	Key	Normal token	Alt token	Ctrl token	Shift token	Num lock
21	[F]	2166	2100	2106	2146	-
22	[G]	2267	2200	2207	2247	-
23	[H]	2368	2300	2308	2348	-
24	[J]	246A	2400	240A	244A	-
25	[K]	256B	2500	250B	254B	-
26	[L]	266C	2600	260C	264C	-
27	[;]	273B	ignore	ignore	273A	-
28	[‘]	2827	ignore	ignore	2822	-
29	[#]	2960	ignore	ignore	297E	-
2A	Left [Shift]	ignore	ignore	ignore	-	-
2B	[\]	2BSC	ignore	2B1C	2B7C	-
2C	[Z]	2C7A	2C00	2C1A	2C5A	-
2D	[X]	2D78	2D00	2D18	2D58	-
2E	[C]	2E63	2E00	2E03	2E43	-
2F	[V]	2F76	2F00	2F16	2F56	-
30	[B]	3062	3000	3002	3042	-
31	[N]	316E	3100	310E	314E	-
32	[M]	326D	3200	320D	324D	-
33	[,]	332C	ignore	ignore	333C	-
34	[.]	342E	ignore	ignore	343E	-
35	[/]	352F	ignore	ignore	353F	-
36	Right [Shift]	ignore	ignore	ignore	-	-
37	[Prtsc]	372A	ignore	7200	Print	-
38	[Alt]	ignore	ignore	-	ignore	-
39	[Space Bar]	3920	3920	3920	3920	-
3A	[Caps Lock]	ignore	ignore	ignore	ignore	-
3B	[F1]	3B00	6800	5E00	5400	-
3C	[F2]	3C00	6900	5F00	5500	-
3D	[F3]	3D00	6A00	6000	5600	-
3E	[F4]	3E00	6B00	6100	5700	-
3F	[F5]	3F00	6C00	6200	5800	-
40	[F6]	4000	6D00	6300	5900	-

Key Code	Key	Normal token	Alt token	Ctrl token	Shift token	Num lock
41	[F7]	4100	6E00	6400	5A00	-
42	[F8]	4200	6F00	6500	5B00	-
43	[F9]	4300	7000	6600	5C00	-
44	[F10]	4400	7100	6700	5D00	-
45	[Num Lock]	ignore	ignore	Pause	ignore	-
46	[Scroll Lock]	ignore	ignore	(Break)	ignore	-
47	Key pad [7]/[Home]	4700	ignore	7700	-	4737
48	Key pad [8]	4800	ignore	ignore	-	4838
49	Key pad [9]/[PgUp]	4900	ignore	8400	-	4939
4A	Key pad [-]	4A2D	ignore	ignore	-	4A2D
4B	Key pad [4]	4B00	ignore	7300	-	4B34
4C	Key pad [5]	ignore	ignore	ignore	-	4C35
4D	Key pad [6]	4D00	ignore	7400	-	4D36
4E	Key pad [+]	4E2B	ignore	ignore	-	4E2B
4F	Key pad [1]/[End]	4F00	ignore	7500	-	4F31
50	Key pad [2]	5000	ignore	ignore	-	5032
51	Key pad [3]/[PgDn]	5100	ignore	7600	-	5133
52	Key pad [0]/[Ins]	5200	ignore	ignore	-	5230
53	Key pad [.]/[Del]	5300	ignore	ignore	-	532E
54 ..7F	Undefined	ignore	ignore	ignore	ignore	ignore

Key tokens described as ‘ignore’ are actually returned with bit 7 set, which cause them to be ignored.

The following key codes are non-standard. They can be used in INT 16H function 24H, to reprogram the extra **Hunter 16** keys and reprogram keys to perform non-standard actions:

Key Code	Key
B1	[F10] ([PAW][0])
B2	[F9] ([PAW][9])
B3	[F8] ([PAW][8])
B4	[F7] ([PAW][7])
BS	[F6] ([PAW][6])
B6	[F5] ([PAW][5])
B7	[F4] ([PAW][4])
B8	[F3] ([PAW][3])
B9	[F2] ([PAW][2])
BA	[F1] ([PAW][1])
BB	Keypad + ([PAW][#])
BC	Keypad - ([PAW][/])
BD	[[] ([PAW] plus keypad [3])
BE	[]] ([PAW] plus keypad [1])
BF	[Scroll Lock] ([PAW]plus keypad [9])
CO	Print Screen ([Shift][PAW] plus keypad [7])
C1 ..E6	Undefined
E7	As EC, but requires [Shift] too
E8	Increase contrast with wraparound ([PAW][E])
E9	Reserved
EA	Reserved
EB	Reserved
EC	Turn off backlight and increase level (see note below)
ED	Momentary SHIFT (shift next key)
EE	HOT key ([PAW][H])
EF	Toggle sound ([PAW][S])

Note: Key code EC set the backlight level to 0, then gradually increases the level until the key is released.

Key Code	Key
F0	Toggle MAP mode ([PAW][M])
F1	Restore window ([PAW] plus keypad [5])
F2	Flip window ([PAW][F])
F3	Shift window right ([PAW][Right])
F4	Shift window left ([PAW][Left])
FS	Shift window down ([PAW][Down])
F6	Shift window up ([PAW][Up])
F7	Reserved
F8	Toggle backlight ([PAW][L])
F9	Decrease backlight level ([PAW][C])
FA	Increase backlight level ([PAW][R])
FB	Decrease contrast ([PAW][X])
FC	Increase contrast ([PAW][E])
FD	[PWR]
FE	[PAW]
FF	No action ie ignore key